

Anthony Rossi

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EDUCATION

California Polytechnic

Bachelor of Science in Computer Science

San Luis Obispo, CA

September. 2022 – June 2026

RELEVANT COURSEWORK(* SOON TO BE COMPLETED)

Data Structures, Systems Programming, Algorithms, Computer Security, Programming Languages, Computer Organization, Object Oriented Programming and Design, Software Engineering, Databases, Software Evaluation, Computer Graphics, Theory of Computation, Networks, Parallel Computing, Web Development, Operating Systems, Binary Exploitation, Distributed Systems*, Network Security*

PROJECTS

Software Renderer | C

January 2025 – Apr 2025

- Built a software renderer to deepen my understanding of computer graphics beyond using high-level APIs like OpenGL.
- Wrote a 3D software renderer entirely from scratch using C, which included **triangle rasterization, Z-buffering, perspective handling, texture mapping, basic shader logic, and camera controls**.
- Successfully rendered textured 3D models with correct depth and perspective entirely without external graphics libraries, gaining a strong understanding of the full graphics pipeline and low-level systems programming.

Cloud Performance Testing | Python, Bash, AWS, Google Cloud, Lambda Labs

March 2025 – May 2025

- Evaluated the performance and cost-efficiency of training **NeRF Studio** models across different cloud platforms, where setup complexity and resource variability made direct comparison difficult
- Configured cloud environments and automated workflows using **Python and Bash scripts** to evaluate performance metrics of NeRF studio model training across multiple cloud infrastructures
- Analyzed computation times, resource utilization, training efficiency, and costs to identify performance bottlenecks and ease of use among different platforms

Arcade Drone Simulation | C++, GLSL, OpenGL

May 2025 – March 2026

- Developed an engaging **drone simulation**, aimed to be more accessible and game-like compared to the current simulations on the market.
- Developed a first-person drone simulator in **C++ using OpenGL and GLSL**, implementing real-time graphics features such as lighting, **procedural shaders, dynamic LOD and custom drone physics**.
- Delivered a smooth, interactive simulation with **real-time trick detection**, introducing a freestyle-oriented system not typically supported in traditional drone simulators.

Raspberry Pi Home Server | C, Linux

January 2026 – Present

- Setup a **secure and self-hosted** solution for file synchronization and remote system access, while working within the constraints of a low-power Raspberry Pi and public internet exposure.
- Built a home server on a Raspberry Pi using Linux, configuring networking and security features including **WireGuard VPN** tunneling and Syncthing for **cross-device file synchronization**.
- Allowed for secure, reliable file syncing, remote workstation access across multiple devices, and host for **custom web server written in C**.

EXPERIENCE

Sprouts Farmers Market - Cashier

June 2025 – Oct 2025

- Worked as a cashier attending to customers' requests, providing good customer service
- Collaborated with team members to maintain store operations during peak hours

TECHNICAL SKILLS

Languages: C/C++, Rust, Java, Python, RISC-V, SQL, JavaScript, HTML/CSS, Racket

Developer Tools: Git, Vim, Linux, AWS, Google Cloud Platform, Lambda Labs

Libraries: OpenGL